

Review

Recommendation networks of homogeneous products on an E-commerce platform: Measurement and competition effects

Zhen Zhu^{a,1,*}, Shan Wang^{b,2}, Fei Wang^{a,3}, Zihang Tu^c

^a The Center for International Cooperation in E-Business, School of Economics and Management, China University of Geosciences, Wuhan, Hubei 430078, PR China

^b Department of Finance and Management Science, University of Saskatchewan, 25 Campus Drive, Saskatoon, SK S7N 5A7, Canada

^c School of Economics and Management, China University of Geosciences, Wuhan, Hubei 430078, PR China

ARTICLE INFO

Keywords:

Recommendation networks
Global-network
Network structural properties
Competition effects
Evenness of distribution
Clustering of structure
Substitution recommendation

ABSTRACT

Extant studies have focused on the sales benefits of product recommendation networks; however, the competition effects of such networks have been overlooked. Understanding the competition effects of product recommendation networks is important within an e-commerce platform because maintaining a proper level of competition is critical to the platform ecosystems. This research adopts a data-driven approach and social network analysis, elaborating upon three structural properties of recommendation networks – *centrality*, *connection*, and *scale* – and operationalizes them as *evenness of distribution*, *clustering of structure*, and *network size* at the global-network level. The competition effects of the three proposed network variables were tested based on a daily panel dataset of 320,981 package tour products from *Tuniu.com*, a top travel service platform in China featuring homogeneous products. Empirical results confirmed that the three proposed structural network properties increase market competition in recommendation networks of homogeneous products. This research contributes to our understanding of the impact of recommendation networks on market competition in the context of homogeneous product markets and offers practical implications of substitution recommendation for platform governance. Platform operators can leverage product recommendation networks to alleviate market competition and encourage the entry of sellers with new, innovative products that contribute to the e-commerce platform ecosystems.

1. Introduction

With the rapid growth of electronic commerce (e-commerce) and social commerce (s-commerce) in digital markets (Handarkho, 2021; Xiao, Wu, Xie, & Hu, 2019; Zhu, Zhao, & Bush, 2020), *recommendation systems* are being used to assist online customers in making appropriate purchase decisions (Adomavicius, Bockstedt, Curley, & Zhang, 2013; Choi, Oh, Kim, & Ryu, 2016; Jing, Jiang, Du, & Sugumaran, 2018; Lin, 2014; Oestreicher-Singer & Sundararajan, 2012a; Oestreicher-Singer & Sundararajan, 2012b; Zhang, Zhao, & Gupta, 2018). E-commerce platforms provide millions of well-designed product webpages to display

information on multiple products that attracts online customers and informs customer decision making (Liu, Zhang, Zhang, & Zhao, 2019; Oestreicher-Singer & Sundararajan, 2012b; Wang, Yu, & Deng, 2019). Therefore, it can create the problem of information overload in e-commerce platforms (Lu, Wu, Mao, Wang, & Zhang, 2015). To address this problem, platforms utilize recommendation systems to generate and display hyperlinks of relevant products on product pages (Adomavicius, et al., 2013), which creates product recommendation networks (Lin & Wang, 2018; Lin, 2014; Oestreicher-Singer, Libai, Sivan, Carmi, & Yassin, 2013; Oestreicher-Singer & Sundararajan, 2012a; Pathak, Garfinkel, Gopal, Venkatesan, & Yin, 2010; Wang, Xu, & Wang, 2018). Amazon.

Abbreviations: *PRVar*, Variance of PageRank values; *AvgCC*, Average clustering coefficient; *IOC*, Intensity of competition; *AvgDemand*, Average the number of purchasing tourists; *AvgPrice*, Average the listed prices; *PriceVar*, Variance the listed prices.

* Corresponding authors at: School of Economics and Management, China University of Geosciences, 68 Jincheng Street, Donghu High-tech Development Zoon, Wuhan, Hubei 430078, PR China.

E-mail addresses: zhuzhen2008@gmail.com, zhuzhen@cug.edu.cn (Z. Zhu), wang@edwards.usask.ca (S. Wang), wangfei086071@163.com (F. Wang), tu_zihang@163.com (Z. Tu).

¹ ORCID: 0000-0001-7735-2192.

² ORCID: 0000-0002-0698-4341.

³ ORCID: 0000-0003-1347-3723.

<https://doi.org/10.1016/j.eswa.2022.117128>

Received 12 September 2021; Received in revised form 22 January 2022; Accepted 28 March 2022

Available online 2 April 2022

0957-4174/© 2022 Elsevier Ltd. All rights reserved.

com is the first e-commerce platform to introduce recommendation networks⁴ titled “customers who bought this item also bought” (Jabr & Zheng, 2014).

Product recommendation networks are a double-edged sword for e-commerce platforms. On the one hand, they improve product sales on platforms because they help users find products that fit their preferences (Zhang & Bockstedt, 2020). A product has higher accessibility if the system recommends it, which leads to higher demand for the product (Adamopoulos, Ghose, & Tuzhilin, 2022). The product recommendation networks also promote co-view and co-purchase, thereby improving the cross-selling of relevant products through a complementary recommendation (Lin, Goh, & Heng, 2017). On the other hand, recommendation networks create competition pressure among sellers. The recommendation links can distract customers, leading customers to switch to a different product. The competition pressure is especially strong in markets with homogeneous products because the recommended products are substitutes. Homogenous products are considered homogenous when they are perfect substitutes, and buyers perceive few actual differences (Khemani & Shapiro, 1993). For example, tour service products with the same destination are such products, which are offered by different vendors but share the same routes and a set of similar hotel chains. In homogeneous product markets, a customer switching to a different product might mean that the customer switches to a substitute product sold by a different seller. Thus, product recommendation systems are likely to re-shape the competition among e-commerce sellers because product recommendation systems have strong substitution effects, which can give rise to the change of competitive dynamics among sellers (Jabr & Zheng, 2014; Yang & Gao, 2017; Zhang & Bockstedt, 2020).

The wide application of recommendation systems on e-commerce platforms increasingly draws interest from researchers (Chen, Lu, Wang, & Pan, 2019; Gorgoglione, Panniello, & Tuzhilin, 2019; Jing, et al., 2018; Shao, Li, & Bian, 2021). There are two dominant streams of research. One stream focuses on the technique and algorithm design. Researchers have either developed new recommendation techniques or improved existing techniques to achieve better recommendation performance (He & Chu, 2010; Pazzani & Billsus, 2007; Schafer, Dan, Herlocker, & Sen, 2007; Yang, Jo, Jeon, Kim, & Kang, 2022; Zhang et al., 2013). The other research stream, which has arisen in recent years, studies the impact of product recommendation networks. For example, some empirical studies have emerged to investigate the economic effects of product recommendation systems on product sales (Carmi, Oestreicher-singe, Stettner, & Sundararajan, 2017; Fleder & Hosanagar, 2009; Oestreicher-Singer, et al., 2013; Oestreicher-Singer & Sundararajan, 2012a; Oestreicher-Singer & Sundararajan, 2012b). Some researchers have also found multiple positive benefits resulting from complementary recommendations (Lin, et al., 2017; Oestreicher-Singer & Sundararajan, 2012b).

However, limited attention has been paid to the competition effect of product recommendation networks by the homogeneous products (Ghasemaghaei, Hassanein, & Benbasat, 2019; Xu, Benbasat, & Cenfetelli, 2014). The right level of competition is critical for an e-commerce platform. High competition erodes the profits of sellers and leads to massive market exits, while low competition and high market dominance deter new entrants into the market. Understanding the impact of product recommendation systems on market competition is paramount for e-commerce platforms ecosystems to create sustainable growth opportunities. Furthermore, product recommendation systems and their resulting product networks are highly manipulable. They can be strategically manipulated to achieve competition governance goals by the platform (Yang & Gao, 2017). Such manipulation potential of product

recommendation systems has been noted by several researchers (Hosanagar, Krishnan, & Ma, 2008; Li, Chen, & Raghunathan, 2018; Yang & Gao, 2017). Product recommendation systems can be an invisible hand for e-commerce platforms to influence competition in platform ecosystems.

Furthermore, the extant literature has investigated the various economic impacts on recommendation networks of search products (e.g., book, camera) (Jing, et al., 2018; Leem & Chun, 2014; Lin, 2014; Liu, et al., 2019; Oestreicher-Singer & Sundararajan, 2012a; Pathak, et al., 2010); Most of the research has ignored the impact on experience products. Search products are products and services whose features and characteristics can be easily evaluated before purchase (Mittra, Reiss, & Capella, 1999; Nelson, 1970), while experience products such as travel service, consulting service, and education are products and services whose crucial attributes can be evaluated only after use (Im, et al., 2016). One can only assess their quality after customers purchase and use them (He, Liu, Tan, & Li, 2021). Therefore, the critical difference between these two product and service categories affects the consumers' perceived uncertainty and search behaviors (Im, Jun, Oh, & Jeong, 2016). Given the high cognitive effort and search costs associated with evaluating the brand names and quality of experience products, consumers' decisions could be more acutely affected by product recommendation networks, including both the sales improvement effect in the case of complementary recommendation and the competition effect in the case of substitutive recommendation. Therefore, future research should examine the effect of recommendation networks for experience products and services on e-commerce platforms.

The present study is motivated by the above-mentioned two research gaps and investigates how product recommendation networks affect market competition in the context of homogenous and experience products. The extant empirical literature related to networks' economic effects is summarized to identify three attributes of networks: *centrality*, *connection*, and *scale*. Using the data-driven methodology, we further propose three network variables *at the global-network level* to operationalize the three attributes of networks and analyze the impact of three network variables on market competition. We collected a daily panel dataset from Tuniu.com, one of the top package tour product e-commerce platforms in China, to estimate our econometric model. Our research offers both important theoretical and practical implications. Theoretically, this research summarizes three attributes to describe the properties of recommendation networks for homogeneous product markets, study their impacts on market competition, and directs researchers' attention to the competition effect of recommendation systems. Practically, it guides practitioners to monitor the market competition and potentially moderate the recommendation systems to maintain healthy competition in the platforms.

The rest of this article is organized as follows. Section 2 reviews the literature related to recommendation techniques and metrics about product recommendation networks. Research data and measurements about structural properties of the recommendation networks *at the global-network level* are introduced in Section 3. Next, a feasible generalized least squares (FGLS) model is used to estimate the econometric model for the competition effect of recommendation networks in Section 4. Finally, we discuss theoretical contributions, management implications, and outline potential directions in Section 5.

2. Related works

2.1. The effect of recommendation systems

A recommendation system is an integration of software modules and hardware aiming to predict the preferences of users (individuals or businesses) by collecting (explicitly or implicitly) and analyzing users' information, items, and interaction between them in order to accurately and timely recommend suitable items to users (Bobadilla, Ortega, Hernandez, & Gutiérrez, 2013; Lu, et al., 2015). In this research,

⁴ We are interested in online product networks that are created by the use of recommendation systems in this paper. Thus, we use the terms *recommendation systems* and *recommendation network(s)* interchangeably.

recommendation networks refer to the directed product networks generated by recommendation systems.

Three categories of techniques are used to develop recommendation systems on e-commerce platforms (Burke, 2002; Lu, et al., 2015; Majd & Hobson, 2020; Xu, et al., 2009) and mobile channel platforms (Adamopoulos, Ghose, & Tuzhilin, 2022). First, content-based techniques recommend items similar to those that a specific user has shown interest in before (such as viewing and clicking) using matching algorithms to match the characteristics of items and user profiles (Abbasi-Moud, Vahdat-Nejad, & Sadri, 2021; Lu, et al., 2015). Second, collaborative filtering-based techniques recommend items to a specific user based on the interests of other similar users by finding correlations among users (Choi, et al., 2016). Third, hybrid techniques take advantage of the above-mentioned two recommendation techniques to overcome the drawback of using a single recommendation approach (Lu, et al., 2015; Zhang, et al., 2013). Among numerous algorithms, the content-based approach, which makes a recommendation based on product similarity (Son & Kim, 2017), has become one of the most popular recommendation approaches that create excellent performance (Abbasi-Moud, Hosseiniabadi, Kelarestaghi, & Eshghi, 2022).

Recommendation systems have produced profound value impacts on the e-commerce platforms and their participants. The business value generated by the presence of recommendation systems is mainly due to the awareness effect and salience effect (Lee & Hosanagar, 2020). Recommendation systems increase the visibility and exposure of a product and reduce consumer search cost (i.e., the awareness effect) and furthermore enhance customer trust towards the products and reduce product fit uncertainty (i.e., the salience effect). A range of business value metrics has been studied, including sales and product demand, product views, click-through rates, conversation rates (Lee & Hosanagar, 2020), customer retention rates (Baum & Spann, 2014; Park & Han, 2013), and consumer perception about products as reflected in customer product preference construction (Adomavicius, et al., 2013; Häubl & Murray, 2003) and online review ratings (Cosley, Lam, Albert, Konstan, & Riedl, 2003; Lin & Wang, 2018). Most of the studies of the business value of recommendation systems are conducted at the individual product or consumer level. Jannach and Jugovac (2019) provide a review of the business values of the recommendation systems.

However, the impact of recommendation systems on market competition is not fully addressed in the literature. Two streams of research seem to suggest that recommendation systems are likely to influence market competition. First, the sales benefits accrued from the use of recommendation systems are not equal for all products. Some products benefit more from the recommendation systems, and that the competition among products can decrease, since the sales will be condensed to some products disproportionately. The relationship between recommendation systems and product sales is controlled by the mix of two major mechanisms: complementary and substitutive mechanisms (Hinz & Eckert, 2010). If the recommended products are complementary, users tend to buy more, so all the products exposed by the recommendation systems benefit from such exposure to a certain degree. If the recommended products are substitute products, a user can easily switch from one product to the other, resulting in either increased or reduced sales for the focal products. Researchers have found that, for a focal product, the positive gain from recommendation systems can be moderated by the competing products' attributes, such as their prices, centrality, and review ratings (Huang, Yang, & Zheng, 2021) as well as consumer attributes such as budget constraints, time constraints, and demand restrictions (Hinz & Eckert, 2010). The nature of the focal product can also moderate the gain from recommendation systems. For example, popular products (Lee & Hosanagar, 2020) and experienced products are found to benefit more from recommendation systems. The strength of the substitutive effect explains the moderating role of those factors.

Second, the product-market-level research about the impact of recommendation systems on sales diversity offers perplexing results on

product competition. On the one hand, researchers find that recommendation systems can enable customers to discover new products, increase sales diversity, and facilitate the long-tail business model (Anderson, 2006; Brynjolfsson, Hu, & Smith, 2006; Jabr & Zheng, 2014; Oestreicher-Singer & Sundararajan, 2012a). Thus, the competition among products increases. On the other hand, some researchers found that recommendation systems reduce sales diversity (Lee & Hosanagar, 2020), since similar users end up exploring the same kinds of products, resulting in the concentration bias at the aggregate level. This concentration bias happens despite the increase in individual-level consumption diversity. The bias is partly due to the endogeneity of the recommendation algorithms used by the system; the recommendation system uses customers' co-purchase data to make recommendations, and new customers follow this pattern to make purchasing decisions and thus strengthen the co-purchase product linkages. According to this stream of research, recommendation systems in fact lead to product dominance and low market competition.

Although the extant literature suggests the potential impact of product recommendation systems and the resulting recommendation networks on market competition, such market competition is neither explicitly studied nor empirically tested. This research intends to fill this research gap.

2.2. The network structure of product recommendation networks

Recent studies have examined the complementary impact of recommendation networks on product demand from the perspective of network structure (Oestreicher-Singer & Sundararajan, 2012b; Stephen & Toubia, 2010). Three levels of networks, i.e., ego-network, dyadic-network, and global-network, have been investigated in previous empirical research (Lin & Wang, 2018).

- Researchers show great interest in examining the influence of recommendation networks on single products at the ego-network level. Oestreicher-Singer and Sundararajan (2012b) studied the role of network position, which is operationalized using the metric of degree centrality and suggested that a more centrally located product will amplify the influence of complementary products on each other's demand. Researchers also separated PageRank centrality and found the difference in the values of incoming and outgoing links in recommendation networks (Oestreicher-Singer, et al., 2013).
- Researchers also studied two connected products in a recommendation network at the dyadic network level and examined their interaction with and influence on each other. Lin and Wang (2018), for example, uncovered the tendency of convergence of Word of Mouth (WOM) rating between products in a recommendation network. WOM rating refers to the rating provided by online consumers.
- At the global-network level, researchers tried to describe the features of the whole recommendation network and studied the economic outcome of these features. For example, Oestreicher-Singer and Sundararajan (2012a) associated the influence of recommendation networks in all book categories on Amazon with the unequal distribution of each category's revenue.

Studying the competition effect of recommendation networks requires the present study to focus on the network structural properties at the global-network level. However, to our best knowledge, no prior literature has proposed metrics systematically to describe features of recommendation networks for homogeneous products. Thus, we need to summarize the existing metrics, which motivates our subsequent review.

2.3. SNA and metrics of the recommendation networks

Many metrics from social network analysis (SNA) were used to measure product recommendation networks. SNA presents social

structure through networks and graph theory (Otte & Rousseau, 2002). This theory conceptualizes social structure as a network consisting of nodes (e.g., individual actors, or people) and the links (relationship or interaction between nodes) that connect them. We summarized the SNA metrics used in previous empirical research that studied the economic effect of product recommendation networks. Three attributes, *centrality*, *connection*, and *scale* (Hanneman & Riddle, 2011; Oestreicher-Singer & Sundararajan, 2012a), were identified to explain the structural properties of product recommendation networks. These attributes shed light on the selection of network structural properties and the subsequent investigation of how these properties affect market competition. More detailed summaries are shown in the Appendix.

The *centrality* attribute is the most widely used approach to understanding individuals' significance relative to their neighbors in social network analysis (Newman, 2010). The core idea of centrality is that an actor is more important if it is more central to the social structure (Hanneman & Riddle, 2011). An actor's central position within a social network significantly impacts its behaviors and those of others. In product recommendation networks, the *centrality* attribute determines the products' accessibility, because a product with high centrality has a more central position to attract the attention of customers. Following Leem and Chun's research, the *centrality* attribute usually includes degree centrality, closeness centrality, betweenness centrality, eigenvector centrality, and PageRank centrality (Leem & Chun, 2014).

The *connection* attribute focuses on how individuals connect with others in social networks. Moreover, the particular connection pattern between actors can significantly affect the behavior of networks (Newman, 2010). The *connection* attribute was used to measure the strength of the relation between two actors or its valences (positive or negative), such as group-external and group-internal ties, in-coming and out-going paths, and strong and weak ties of neighborhoods (Hanneman & Riddle, 2011). In product recommendation networks, the direction of a connection (from A to B, or B to A) determines the flow of consumer attention and traffic to a product. Attribute assortative mixing and clustering coefficients were widely used in previous research to measure the *connection* attribute of product recommendation networks (Goldenberg, Oestreicher-Singer, & Reichman, 2012; Oestreicher-Singer & Sundararajan, 2012b; Stephen & Toubia, 2010).

As the third dimension, the *scale* of recommendation networks was used to describe the size attribute of a network measured by the number of nodes in this network. Individuals in a social network usually have limited capacities to build and maintain a relationship with each other. While the scale of the network is growing, it is extremely difficult for each individual to remain connected with others (Hanneman & Riddle, 2011).

3. Data collection and metrics

3.1. Data collection and processing

As a type of experience good, package tour services were increasingly popular among global tourists shopping on e-commerce platforms before Covid-19. A package tour service is a tour package (hereafter "tour package products") in which services such as transport, accommodations, and guides are sold together by a vendor known as a tour operator. We obtained daily panel data from Tuniu.com, one of China's top e-commerce platforms that sell tour package products. More than one million products are available on Tuniu, accounting for 28.63% of China's online travel market share in 2018 (AnalysisConsulting, 2019). Every year, more than 15 million tourists purchase packaged tour products online on this platform.

Like Amazon, Tuniu designates a web page for each tour package product to display its primary attributes, including the product ID, product title, price, promotion, and detailed service included in the package such as itinerary, quality of hotels, meals, and airport pickup. In addition, a set of high-performance recommendation systems supports

intelligent recommendations for each product (see Fig. 1).

Tuniu's recommendation systems has been widely recognized and examined in previous traveling recommendation research (Chen, Cao, Wang, Liang, & Zhu, 2022). Unlike other e-commerce platforms that use several recommendation approaches (e.g., Tmall, Amazon), Tuniu only provides a simplified content-based approach to recommend products. In other words, for each product, the recommendation systems of Tuniu identify the critical attributes of products (e.g., destination, service, and price) and provide links to only the most similar products as recommendations for focal products. The strategic motivation of the platform is to promote continued innovation of products from travel agencies. Recommendations with similar products motivate travel agencies to develop more and better differentiated products on the e-commerce platform because of the market competition brought by recommendation systems. Therefore, products available in Tuniu's recommendation networks are *substitutes* for and *competitors* of each other. Under this condition, product competition accrued by recommendations systems can foster service innovation, which adds to the choices customers have when they visit and search for a potential product.

Moreover, the recommendation algorithm of Tuniu does not analyze customers' online behavior data to provide personal recommendations, similar to standard content-based algorithms. The recommendations on each product webpage are the same for different customers, and customer heterogeneity is ignored. Therefore, recommendation networks from Tuniu provide us an excellent opportunity to empirically test the relationship between recommendation network structures and the intensity of competition of package tour products.

We collected daily webpage snapshots of products, including data on all product attributes and the list of recommended products for each product. Fig. 2 shows the data collection and preparation process. First, we selected both domestic departure cities and outbound destinations as the markets for this research. On the Tuniu platform, 12 top popular outbound destinations and 16 top popular domestic departure cities were chosen as search conditions to create a sample frame of products. The demands for products originating from or bound to the cities account for 92% of China's online travel market share in 2018. The search engine was used to retrieve all products available on Tuniu (see Fig. 3).

Second, we scraped and extracted product data from Tuniu. A Python program was developed to collect product attributes, descriptions, and recommendation lists of all products (please refer to Table 1 for a list of data items collected). We collected daily data from June 1st to June 30th, 2018. Only package tour products⁵ in the search results were collected for empirical analysis.

Third, because the research focuses on outbound products (i.e., international travel products), we extracted data related to 12 international destinations departing from major domestic cities with international flights. A total of 320,981 products over 30 days were retained for the study. Each international destination is considered a travel market, so the data includes 12 markets. In all 12 markets, the average selling quantity of tour package products was 135, and the average price was 10,636 Yuan (in RMB).

Finally, we derived the edge list from the dataset and calculated the critical metrics that measure the structural features of product recommendation networks. Gephi, an open-source network analysis software written in Java, was used to calculate the values of independent variables, as described in Fig. 2. The next sub-section will introduce the metrics to measure independent variables (IV) and dependent variables (DV).

⁵ There are several types of products on Tuniu. We only focus on outbound products, which are the main source of revenue for this e-commerce platform.



Fig. 1. Product page within the recommendation systems on Tuniu.

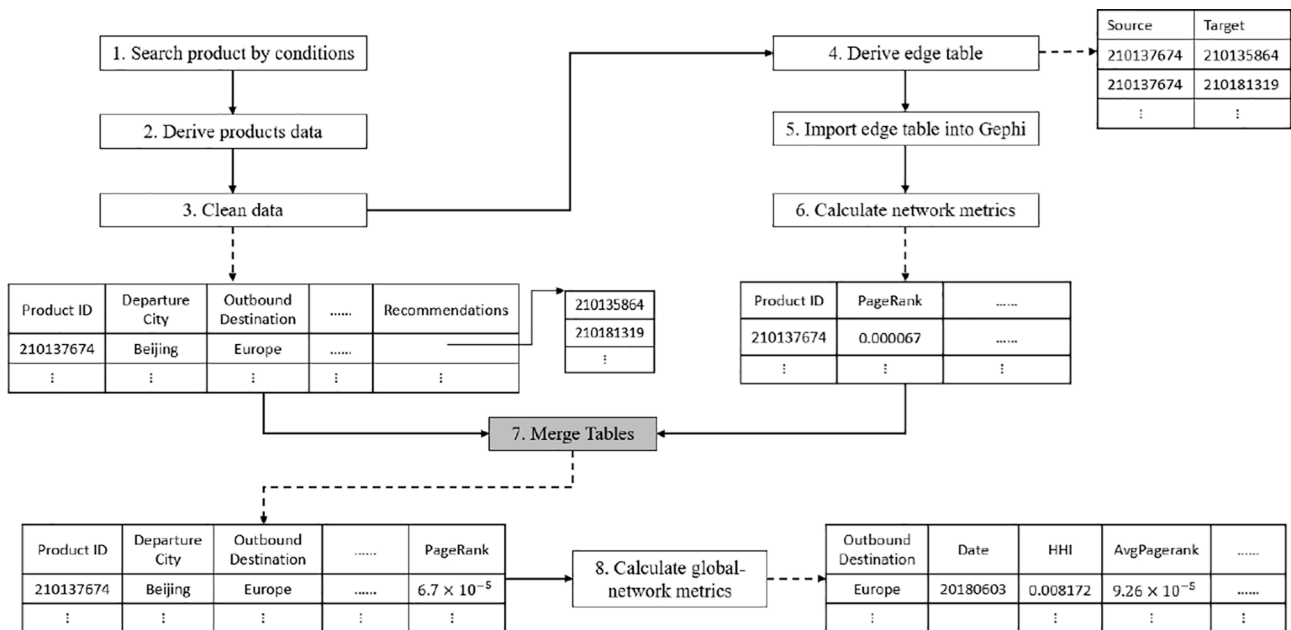


Fig. 2. Data processing scheme.

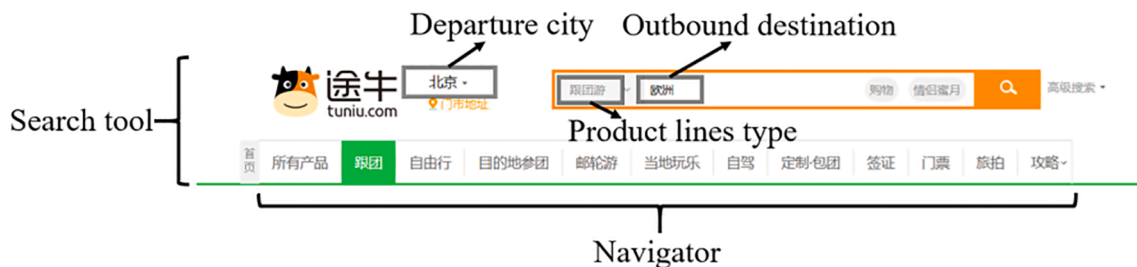


Fig. 3. Search engine on Tuniu.

Table 1

Data explanation.

Data	Explanation	Case Example
Departure city	The domestic city of products.	Beijing
Outbound destination	The outbound destination of products.	Europe
Date	Date of Data collection	2018-06-01
Product ID	ID of every product in recommendation networks	210137674
Product type	Type of products	Package tour
Price (RMB yuan)	The price of products sold on Tuniu that day	15999.00
Tourist Number	Number of purchasing tourists	103
Recommended product IDs	A list of products (represented by their ID numbers) recommended by recommendation systems	210135864 210181319 210181797 210136255

3.2. Metrics and variables

3.2.1. Structural properties of product recommendation networks

Following the above literature review (Oestreicher-Singer & Sundararajan, 2012a; Stephen & Toubia, 2010), we propose measurement variables to analyze the structural properties of product recommendation networks at the global-network level.

In global network analysis, the scale and connection were identified as two critical attributes in analyzing the structure properties (Hanneman & Riddle, 2011; Oestreicher-Singer & Sundararajan, 2012a). The scale of a recommendation network is measured by *size of the networks*. The *size of the networks* should be considered when analyzing the network structure's structural properties. In product recommendation networks, each product's capacity for linking to others is strictly limited by the computing resource provided by e-commerce platforms at a given period. For example, a directed network with only ten nodes uses 90 links to connect all nodes in both directions. However, this process takes 9900 links in a network with 100 nodes. The total number of links grows almost exponentially (i.e., $n(n-1)$, where n is the total number of nodes). An increase in the number of products will quickly undermine the capacity of maintaining the relationship with one another and further affect the structure of recommendation networks. If new products are added to the recommendation network in a market, traffic will be shared within these products with a fixed number of network connections, which is likely to lead to intensified competition in the market.

Based on graph theory, while the *size* is a scale property of the node set, the *clustering of structure* reflects the *connection* property of networks between these nodes (Hanneman & Riddle, 2011). Clustering affects the speed at which information diffuses in a network. The closer the nodes, the faster the information can spread in a network. The recommendation networks with higher clustering enable customers to find products more efficiently, leading to intensified competition, since customers can easily switch to other products under this low searching cost.

However, the two attributes mentioned above are not sufficient to measure the *centrality* dimension of networks from the global-network level perspective. Stephen and Toubia (2010) suggest that centrality distribution equality is another critical metric to view the centrality attribute of network structure. We propose the concept of *evenness of distribution* to describe the centrality distribution equality of each product recommendation network in a certain market. Unlike other social networks, recommendation networks transfer customer attention (or traffic) from one product page to another. In addition, the traffic that a product receives from a recommendation network directly determines the benefit that products get from the network. Therefore, a recommendation network with high *evenness of distribution* tends to guide customers to visit every product (network flow is equally distributed

among all products), while other recommendation networks with low evenness prefer to guide them to visit a few products (only a few products share network flow) (See Fig. 4. for an illustration of the concept). There are two networks presented in Fig. 4. Network 1 is highly centralized, and Network 2 is even in terms of the distribution of centralities. In centralized Network 1, the recommendation network directs all traffic to the central node A, but in the equally distributed Network 2, the traffic is directed to all the nodes equally. Therefore, the *evenness of distribution* of the recommendation network is associated with global competition, and a concentrated network such as Network 1 is likely to foster monopolies or big sellers, thus reducing competition in the market.

3.2.2. Independent variables (IV)

1. *PRVar*: The *evenness of distribution* of product recommendation networks is measured by the variance in PageRank values (*PRVar*) across products in a certain market.

PageRank (*PR*) is an iterative website-ranking algorithm used by Google Search for search optimization (Brin, Page, Motwani, & Winograd, 1999; Oestreicher-Singer, et al., 2013). It is considered the most popular centrality algorithm to calculate the influence of webpages, because it is the only one that evaluates the importance of web pages by simulating users' actual browsing behavior (Oestreicher-Singer & Sundararajan, 2012a; Oestreicher-Singer & Sundararajan, 2012b). Assume that a recommendation system generates recommendation networks consisting of nodes (i.e., product or product pages) and links among nodes (i.e., hyperlinks), then the PageRank of a product in recommendation networks can be calculated iteratively:

$$PR_i = \frac{1 - \alpha}{n} + \alpha \sum_{j \in In_i} \frac{PR_j}{OutDC_j} \quad (1)$$

where PR_i is the PageRank of node i , and n is the number of nodes in graph G ; α is the damping constant, which is usually set to 0.85; $j \in In_i$ if node j points to node i , and $OutDC_j$ is the total number of links originating from product j . The program was initialized with every node having equal probability and terminated when the sum of the distances between the two successive PageRank values of all nodes in the network is less than the preset error tolerance level (Franceschet, 2011), as shown in (2):

$$|PR(t+1) - PR(t)| < \varepsilon \quad (2)$$

where $PR(t+1)$ and $PR(t)$ is a vector of the *PR* values after $t+1$ and t times of iterations, respectively, and ε is the preset tolerance level, which is set to 1×10^{-9} in our study. 1×10^{-9} is considered a threshold value that is sufficient to ensure the accuracy of the calculation, since the ε value normally used by e-commerce platforms is 1×10^{-6} .

We use *PRVar* to assess the evenness of the recommendation network centrality distribution in a destination market. *PRVar* is defined as.

$$PRVar = \frac{\sum_i (PR_i - \overline{PR})^2}{n - 1} \quad (3)$$

where $\overline{PR} = \frac{\sum_i PR_i}{n-1}$; n is the number of nodes in the graph G . It is reasonable to use variance to measure the evenness of distribution, since variance statistically measures the spreadness of the data. A market with a high *PRVar* value means that the traffic from the recommendation networks is distributed to each product unevenly. In other words, the higher the *PRVar* value, the more uneven the network, and vice versa.

2. *AvgCC*: The *clustering of structure* of the entire product recommendation network in a certain market is measured by the average clustering coefficient (*AvgCC*) of all products in this market. Clustering coefficient refers to the ratio of the actual number of links among the nodes to the maximum number of links in the neighborhood of a node. The *AvgCC* can be further used to measure the average clustering

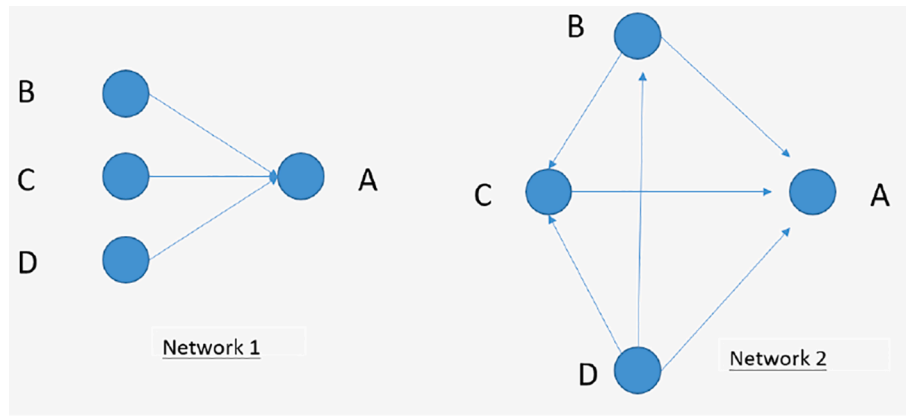


Fig. 4. An illustration of the role of product recommendation network evenness.

coefficients of all the nodes in a whole network. Therefore, the *AvgCC* is defined as.

$$AvgCC = \frac{1}{n-1} \sum_{i=1}^n \frac{|\{e_{jk}|j, k \in N_i, e_{jk} \in E\}|}{|N_i|(|N_i| - 1)} \quad (4)$$

where E is the set of links in graph G ; e_{jk} is one link from node j to node k ; n is the number of nodes in graph G ; N_i is the neighbors of node i , including nodes that directly point to i and nodes that are pointed to by i ; and $|N_i|$ represents the number of nodes in N_i .

3. *Size*: The size of a product recommendation network in a certain market is the total number of products in the network. The formula of scale is defined as.

$$Scale = |V| \equiv n \quad (5)$$

where V is a set of nodes, and n is the number of nodes in graph G .

3.2.3. Dependent variable (DV)

The Herfindahl-Hirschman index (*HHI*) was used to measure the intensity of competition (*IOC*) in a market. *HHI* is the sum of squares of the market share of each product. In our research, *HHI* is calculated as follows:

$$HHI_m = \sum_{k=1}^{Size_m} \left(\frac{TN_{mk}}{\sum_{k=1}^{Size_m} TN_{mk}} \right)^2 \quad (6)$$

where $Size_m$ is the quantity of products in markets m . TN_{mk} is the number of tourists who purchase the product k in this market m , and *HHI* is a decimal number ranging from 0 to 1. The value of *HHI* is affected by market size and sales distribution. An increase in the number of products in a market leads to an increase in competition and a decrease in the market's *HHI*. Also, the evenly distributed sales of products suggest fierce product competition in the market. A lower level of *HHI* suggests a higher degree of *IOC* resulting from product recommendation networks. To facilitate interpretation, we let $IOC = 1 - HHI$ (as 1 is the maximum *HHI* value). Therefore, a higher value of *IOC* indicates a higher intensity of competition.

3.2.4. Control variables

We included three variables to control for the market-specific effects: 1) average number of purchasing tourists (*AvgDemand*) per product in markets, 2) average listed price (*AvgPrice*) in markets, and 3) the variance of the listed prices (*PriceVar*). The *AvgDemand* was included

because the market demand significantly impacts the intensity of competition of products. For each market, we used the average number of purchasing tourists per product to measure market demand.⁶ *AvgPrice* and *PriceVar* were included because the product price influences the intensity of competition in a certain market. The high average market prices can attract the entry of substitute products, therefore intensifying the market competition, whereas evenly distributed prices (i.e., low *PriceVar*) can dampen competition because the market with evenly distributed prices has less attraction for sellers due to limited room for charging a price premium. The abbreviations of all variables used in the research model are shown in Table 2.

4. Econometric model and results

4.1. Econometric model

Econometric problems of groupwise heteroskedasticity and cross-sectional correlation can arise due to the nature of the cross-sectional time-series dataset. According to the Wald test, the null hypothesis that the errors are not heteroskedastic can be rejected ($\chi^2 = 296.89$, $p < 0.001$). The Breusch-Pagan LM test shows cross-sectional correlation problems in our panel dataset ($\chi^2 = 877.20$, $p < 0.001$). Furthermore, the Wooldridge test for autocorrelation (Wooldridge, 2010) indicates the presence of first-order autocorrelation (AR1) in our panel data set ($\chi^2 = 87.31$, $p < 0.001$). In the presence of heteroskedasticity and autocorrelation, an ordinary least squares (OLS) regression is unbiased and consistent, but not efficient, and the standard errors are not correct. Therefore, we use feasible generalized least squares (FGLS) to make the appropriate corrections (Wooldridge, 2010).

We model the impact of structural properties of product recommendation networks on the market competition intensity. The panel-level linear model is specified in the following equation:

$$\begin{aligned} LnIOC_{it+1} = & \alpha + \beta_1 LnAvgDemand_{it} + \beta_2 LnAvgPrice_{it} \\ & + \beta_3 LnPriceVar_{it} + \beta_4 LnPRVar_{it} + \beta_5 LnAvgCC_{it} \\ & + \beta_6 LnSize_{it} + T_t + \alpha_i + \varepsilon_{it} \end{aligned} \quad (7)$$

where $\langle i, t \rangle$ represents market-day combination, α is a constant, β_{1-6} are the coefficients, T_t captures unobserved time-specific effect, α_i is the unobserved time-invariant individual market (i.e., network) effect, and ε_{it} is the error term. The dependent variable and independent variables were specified in logarithmic forms, since the variables are highly skewed. *One-day lagged* independent variables were used to capture the

⁶ We use the average tourist number of products instead of the total number due to the potential risk of multicollinearity between market demand and intensity of competition of recommendation networks.

Table 2
Abbreviations of all variables.

Abbreviations	Attributes of recommendation networks	Definitions	Formulas
AvgCC	Connection	The average clustering coefficient	$\text{AvgCC} = \frac{1}{n-1} \sum_{i=1}^n \frac{ \{e_{jk} j, k \in N_i, e_{jk} \in E\} }{ N_i (N_i - 1)}$
PRVar	Centrality	The variance in PageRank value	$\text{PRVar} = \frac{\sum_{i=1}^n (\text{PR}_i - \overline{\text{PR}})^2}{n-1}$
Size	Scale	The number of nodes in a network	$\text{Scale} = V \equiv n$
AvgDemand	—	The average number of purchasing tourists	$\text{AvgDemand} = \frac{\sum_{i=1}^n \text{TN}_i}{n-1}$
AvgPrice	—	Average product price	$\text{AvgPrice} = \frac{\sum_{i=1}^n \text{price}_i}{n-1}$
PriceVar	—	Product price variance	$\text{PriceVar} = \frac{\sum_{i=1}^n (\text{price}_i - \overline{\text{price}})^2}{n-1}$
IOC	Competition effect	The intensity of competition	$\text{IOC} = 1 - \sum_{k=1}^{\text{Size}_m} \left(\frac{\text{TN}_{mk}}{\sum_{k=1}^{\text{Size}_m} \text{TN}_{mk}} \right)^2$

latency of the competition effect when the markets respond to the changes in the structural properties of recommendation networks and control variables. We estimated this equation using the network structure data over a period of 30 days.

4.2. Model estimation and research results

The descriptive statistics and correlation matrix are shown in Table 3. Most of the absolute values of the correlation coefficients among variables are less than 0.6 (see Table 3). We further computed the variance inflation factors (VIF) to test for any possible multicollinearity. The VIFs for all independent variables in our models are all less than 10, with a mean value of 4.32. According to (Hair, Anderson, Tatham, & Black, 2010), the maximum acceptable level of VIF is 10, suggesting that multicollinearity is not a strong concern in our analysis.

We ran four separate FGLS regression models, as shown in Table 4. Model 1 includes only the control variables related to market demand and prices. All of the control variables have a significant effect on the IOC of product recommendation networks. The results show that both the average number of purchasing tourists (*LnAvgDemand*) per product and the variance of the listed prices (*LnPriceVar*) are significant and negative, suggesting that the average number of purchasing tourists and the variance of the listed prices alleviate the intensity of competition. The average listed price (*LnAvgPrice*) in markets is significant and positive ($\beta_3 = 0.116, p < 0.05$), suggesting that the average listed price impacts market competition in a positive direction. The results suggest that market demand and product price in a platform play a role in influencing market competition.

Model 2 extends the control model (i.e., Model 1) to include the *evenness of centrality distribution* of product recommendation networks that measures the centrality attribute of the networks. Our result shows that the coefficient of *LnPRVar* is negative and significant ($\beta_4 = -0.007, p < 0.01$), indicating that an increase in the uneven structure of recommendation networks is associated with a decrease in IOC. In other words, the inequality of traffic distribution in product recommendation networks will decrease the competition in the destination market. Let's consider an illustrative example of two destination markets: *market A*, where all products have the same PageRank, and *market B*, where only a

few products have high PageRanks, and other products have low PageRanks. One would expect *market B* to be more likely to become a monopolistic market with much less competition than *market A* (Oestreicher-Singer & Sundararajan, 2012a). Most of the network traffic in *market B* is directed to a few popular products, thus leading to the widening sales gap among products and the decrease of the IOC in the market. In contrast, all products in *market A* receive relatively equal traffics from the recommendation network, which leads to an increase of IOC in the market.

Model 3 extends model 2 to include the average *clustering of structure* of the entire product recommendation network that measures the connection attribute of the networks. Table 4 shows that *LnAvgCC* has a positive and significant effect ($\beta_5 = 0.197, p < 0.01$) on the IOC of product recommendation networks. The *AvgCC* refers to the closeness of products to their neighboring products in recommendation networks. A market with closer connections will have a stronger IOC. A possible explanation is that a product with a higher cluster coefficient in the recommendation networks might have more neighbors with competitive property, which is especially true in markets with homogeneous products. The product recommendation networks would easily divert the traffic away from focal products to the competing neighbors due to the connection closeness of the networks. In a destination market with a high CC of recommendation network, traffic can be easily shared among products that contribute to IOC.

Model 4 further extends model 3 to include the *size of the network* that measures the scale attribute of the networks to run the complete model with all of the main effects. Our result shows that *size* is positive and significant ($\beta_6 = 0.291, p < 0.01$), which confirms the positive effect of the *size of the network on market competition*. When more products join the recommendation networks, more rivals exist in the network of recommendation systems, and the competition becomes more intense.

4.3. Robustness test

A variety of additional tests were conducted to test the robustness of our empirical results. First, the estimated effects of the structural properties of product recommendation networks might be biased by the possible existence of unobserved disturbance of time factors. The

Table 3
Description statistics and correlation matrix.

Variables	Mean	S.D.	1	2	3	4	5	6	7
1. LnIOC	3.805	0.790	1						
2. LnPRVar	-18.59	1.63	0.39***	1					
3. LnAvgCC	-0.975	0.266	0.36***	0.56***	1				
4. LnScale	6.520	0.728	0.76***	0.69***	0.68***	1			
5. LnAvgDemand	4.548	0.702	0.74***	0.12***	-0.13***	0.44***	1		
6. LnAvgPrice	8.894	0.709	0.29***	-0.07***	0.40***	0.21***	-0.15***	1	
7. LnPriceVar	16.80	1.96	0.34***	-0.02***	0.40***	0.31***	-0.10***	0.89***	1

Table 4
Results of FGLS and PCSE models.

Variables	FGLS					PCSE
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	3.87*** (0.509)	3.510*** (0.484)	3.447*** (0.443)	0.141*** (0.596)	2.984*** (0.342)	-4.789*** (1.08)
Control Variables						
1. Market Demand						
<i>LnAvgDemand</i>	-0.086*** (0.029)	-0.085*** (0.029)	-0.046** (0.027)	0.102*** (0.039)	0.018 (0.021)	0.417*** (0.065)
2. Price						
<i>LnAvgPrice</i>	0.116** (0.058)	0.140** (0.055)	0.141*** (0.053)	0.198*** (0.037)	0.265*** (0.027)	0.216*** (0.062)
<i>LnPriceVar</i>	-0.012** (0.005)	-0.013** (0.005)	-0.013** (0.005)	-0.010* (0.005)	-0.030*** (0.003)	-0.003 (0.009)
Dependent Variables						
<i>LnPRVar</i>		-0.007*** (0.002)	-0.010*** (0.002)	-0.009*** (0.001)	-0.005** (0.002)	-0.006** (0.003)
<i>LnAvgCC</i>			0.179*** (0.021)	0.134*** (0.018)	0.108*** (0.003)	0.091*** (0.039)
<i>LnSize</i>				0.291*** (0.050)	0.080*** (0.022)	0.255*** (0.097)
Observations	348	348	348	348	720	348
Groups	12	12	12	12	12	12
Periods	29	29	29	29	60	29
Wald chi2	341730	378658	286901	2878257	1013224	—
<i>R</i> ²	—	—	—	—	—	0.99

Note: ** $p < 0.01$, * $p < 0.05$, $\psi p < 0.1$.

concern is that seasonality could influence the impact of product recommendation networks for package tour products. Three more daily panel datasets were collected in September 2018, December 2018, and March 2019 to estimate the econometric model. All FGLS results are consistent with our main model estimates (with the dataset of June 2018). We further checked the robustness of our findings using different time frames. A two-month time frame (June and July 2018) was used instead of the one-month (June 2018) time frame to estimate the coefficients of the variables. The estimates are reported in Model 5 in Table 4, and the results are consistent with those estimated using the one-month time frame.

Finally, in the presence of heteroskedasticity and autocorrelation, both FGLS and OLS with panel-corrected standard errors (OLS-PCSE) are widely used for model estimation (Han, Kauffman, & Nault, 2011; Ren & Dewan, 2015). The FGLS approach underestimates the variability of the standard errors of the estimated coefficients, which could lead to overconfidence. Therefore, OLS-PCSE, which retains the OLS parameter estimates but replaces the OLS standard errors with panel-corrected standard errors, is used to check the robustness of our findings. Model 6 in Table 4 shows the results. In Model 6, the coefficients of all three structural properties of product recommendation networks are statistically significant, and the results in the OLS-PCSE model are broadly consistent with those in the FGLS model.

5. Discussion and contribution

5.1. Major findings

Despite the fact that some studies have described the properties of recommendation networks and investigated the economic impacts of the complementary product recommendation (Jabr & Zheng, 2014; Lin, et al., 2017; Oestreicher-Singer & Sundararajan, 2012b; Pathak, et al., 2010), the competition effect of the product recommendation network in homogeneous product markets remains unclear in the literature. In this study, three critical measurements were developed to analyze the structural properties of recommendation networks from a global-network level. A new metric, the variance in PageRank values (*PRVar*) was developed to measure the *evenness of distribution*. The higher the *PRVar* value, the more uneven the product recommendation networks. For the *clustering of structure*, *AvgCC* was developed to describe the average level of all CCs in a product recommendation network. Finally, the *size of the network* was used to measure the *scale* attribute of product recommendation network, similar to the research by Oestreicher-Singer and Sundararajan (2012a).

By estimating the econometric models, the competition effects using the above three measurements of product recommendation network properties were confirmed using a daily panel dataset of homogeneous package tour products from Tuniu. The major findings are summarized in Table 5. Results of the study show that *PRVar* of product recommendation networks is negatively related to the *IOC*. The coefficient of *PRVar* is -0.007 , indicating that a 1% increase in the *PRVar* of the recommendation networks is associated with a 0.007% decrease in the intensity of competition among products. This result highlights the importance of *evenness of distribution* of production recommendation networks in influencing market competition.

The results also suggest that the *AvgCC* of product recommendation networks is significant and positively related to the *IOC* in the destination market, indicating the positive impact of *clustering of structure* on *IOC*. A product seems to enjoy less competition when it is not connected with those products that are highly similar. The coefficient of *AvgCC* (0.179) indicates that a 1% *AvgCC* of product recommendation networks is associated with a 0.179% increase in *IOC*.

We further found that the *size of the network* has a significant and positive effect on *IOC*. The coefficient of *size* (0.291) suggests that a 1% increase in *size* is associated with a 0.291% increase in the *IOC* among

Table 5
A Summary of Major Findings.

Network variables (Network structural properties)	Major findings supported by the empirical models in Table 4
<i>PRVar</i> (network evenness)	Model 2 indicates that an increase in the uneven structure of recommendation networks is associated with a decrease in <i>IOC</i> ($\beta_4 = -0.007$, $p < 0.01$). In other words, the inequality of network centrality distribution in product recommendation networks will decrease the competition in the destination market.
<i>AvgCC</i> (network connection)	Model 3 shows that <i>LnAvgCC</i> has a positive and significant effect on the <i>IOC</i> accrued by product recommendation networks ($\beta_5 = 0.197$, $p < 0.01$). A product recommendation network with closer connections leads to more intense market competition.
<i>Size</i> (network scale)	Model 4 shows that the effect of <i>size</i> (measured by the total number of products in the network) on <i>IOC</i> is positive and significant ($\beta_6 = 0.291$, $p < 0.01$). This confirms that the more products in the recommendation networks, the fiercer the market competition. Many rivals in the recommendation network intensify market competition.

products. This result indicates that extending the scale of the product network, such as launching more products before holidays, will bring a more vigorous intensity of competition in the product market.

Among three structural attributes of product recommendation networks, *size* plays a more salient role in influencing the *IOC* than *AvgCC* and *PRVar*. The magnitude of the coefficient for *size* is much higher than those of *AvgCC* (0.917) and *PRVar* (-0.007).

5.2. Theoretical contributions

Theoretically, this study makes contributions to the emerging literature primarily on the competition effects of recommendation networks.

First, this study makes review contributions primarily to the emerging literature on recommendation networks by offering three dimensions of measurements (see Appendix) to describe the network structure while there is still a scarcity of systematic network measurements. From a global-network perspective (Lin & Wang, 2018), we addressed a research gap in the product recommendation network literature (Jabr & Zheng, 2014; Oestreicher-Singer & Sundararajan, 2012a; Pathak, et al., 2010) and proposed three new measurement variables that operationalize the structural properties of recommendation networks. This study builds on the network theory (Hanneman & Riddle, 2011) and suggests *centrality*, *connection*, and *scale* as the critical attributes to reveal the potential characteristics of product recommendation network structure.

Second, our research extends the previous research focusing on recommendation networks' demand effect (Stephen & Toubia, 2010) and investigates the competition effect of product recommendation network. While previous research has identified the benefits of the recommendation networks, such as co-view and co-purchase through a complementary recommendation (Lin, et al., 2017; Oestreicher-Singer & Sundararajan, 2012b), our *global-network analysis* reveals that recommendation networks can influence the competition effect in a homogeneous product market. In summary, this research sheds new insights and contributes to the product recommendation network research by studying its competition effect.

Third, our results demonstrate a clear understanding of the extent to which attributes of recommendation networks impact market competition in a homogeneous product market. Results demonstrate that the marginal effect of the *size of the network* is nearly 1.5 times higher than the *AvgCC* and 28 times higher than *PRVar* in boosting market competition. While previous research found that recommendation networks positively impact sales through complementary recommendation (Lin & Wang, 2018; Oestreicher-Singer & Sundararajan, 2012b; Stephen & Toubia, 2010), our finding suggests that recommendation networks can instead pose a threat to demand in homogeneous product markets.

Finally, our research is one of the rare studies examining the competition impact of product recommendation networks for experience products and services on e-commerce platforms. Prior research used samples of search products (e.g., book, camera) on either Amazon or Tmall to conduct empirical analysis (e.g., Lin & Wang, 2018; Oestreicher-Singer & Sundararajan, 2012b). Our empirical analysis focuses on package tour products and provides evidence to support the competition effects accrued by recommendation networks of homogeneous products. More importantly, differing from a positive effect in the "long tail" of demand and revenue, we further find that the *size of the networks* also has a positive effect on the *IOC* in interconnected products. The findings of this research thus complement and enrich prior research in providing a more comprehensive understanding of the economic effect of product networks.

5.3. Practical implications

Our study provides important practical implications for e-commerce platforms in governing platform competition using recommendation systems and for platform sellers in managing a product strategy that

responds to the competition effect of recommendation systems.

First, this study offers implications for manipulating product recommendation systems to monitor and influence homogeneous product market competition. An e-commerce platform can be informed that a recommendation network might not always benefit product selling. Competition dynamics can be changed when the structural properties of product recommendation networks changes. Therefore, changed competition dynamics can hurt some product sales in favor of others. Platforms should avoid both over-competition and lack of competition in the product market by monitoring the *IOC* index as proposed in this research, simulating product recommendation network structural attribute values to understand the impact of such adjustment on market competition, and taking action to make the actual adjustments to fine-tune market competition. A platform should leverage the property that production recommendation networks are highly manipulatable (i.e., adjusting recommendation algorithms) to achieve such a goal. For example, maintaining a limited number of recommendations for every product (e.g., three or four products) or reducing the clustering of the structure can help to establish a healthy competition ecosystem on an e-commerce platform. Since the competition effect is observed in the homogenous product market in this research, platforms can also encourage product innovation and differentiation, measure the similarity of products before they are linked and displayed, and carefully select less similar products to display in the recommendation list to avoid over-competition. Overall, the governing platform competition will help sellers to build new digital operations capabilities (Zhu, Lin, Jiang, & Liu, 2022) with consideration of competition dynamics in the product recommendation networks.

Second, the competition effect occurred by homogeneous product recommendation networks can motivate travel agencies to innovate and differentiate products. Travel agents can use product innovation and differentiation to combat the adverse consequences brought by the competition effect. Furthermore, platform sellers should devise an effective marketing strategy for new product launches in a certain market to counter the negative competition effect brought by the recommendation network of homogeneous products. The strategy of launching a large number of homogenous products may be ineffective. When sellers launch too many products in a market, they fail to achieve high sales performance through the recommendation network because of increased competition among the homogeneous products. Therefore, it will encourage the entry of sellers with new, innovative products that contribute to the e-commerce platform ecosystems.

5.4. Limitations and future research

Our study has some limitations, and further research could be done to fill the gap. First, the field data used for this research was from a specific travel service e-commerce platform. Future research could consider further validating the competition effect of homogeneous experience product recommendation networks by examining other e-commerce platforms (e.g., online finance products) and software platforms (e.g., APP platforms). The competition effects on these emerging platforms are worth exploring.

Second, our study focuses primarily on exploring the phenomenon of recommendation networks of homogeneous experience products and services without conceptualizing them from a theoretical perspective. Although the big data methodology and SNA support the econometric model, future research could adopt theoretical perspectives to explain how both demand and competition affect online product sales.

Third, its role in a double-sided platform can be studied to advance the existing research that has focused on one side of platforms only (i.e., the demand-side product sales) (Lin, et al., 2017; Oestreicher-Singer, et al., 2013; Pathak, et al., 2010). Future research can study the role of recommendation systems on both supply-side product innovation and demand-side product sales.

Finally, the focus of our study is the impact of product

recommendation networks on market competition. Future research can further explore the impact of market competition on user decision-making and user welfare. Such studies will have important managerial implications, since the platforms are double-sided platforms that need to manage not only competition among vendors but also user participation.

6. Conclusions

With the wide application of recommendation systems in e-commerce platforms, extant studies focus on the benefit of the recommendation networks without considering the competition effects (Lin & Wang, 2018; Oestreicher-Singer & Sundararajan, 2012b; Park & Han, 2013). Our study summarizes three attributes to describe the properties of recommendation networks with homogeneous products. A set of daily panel data from Tuniu.com was collected to estimate our econometric model for examining the competition effects at the *global-network level*. Results provide empirical evidence that *evenness of distribution*, *clustering of structure*, and the *size of the network* might lead to a high intensity of competition in product recommendation networks. Our research sheds new insights on and contributes to product recommendation networks research by uncovering the competition effect through a data-driven

approach, calling for more attention to the competition accrued by recommendation systems.

CRedit authorship contribution statement

Zhen Zhu: Conceptualization, Methodology, Discussion. **Shan Wang:** Theory, Writing – review & editing. **Fei Wang:** Literature Review, Data Collection. **Zihang Tu:** Methodology, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Authors acknowledge financial support from National Natural Science Foundation of China under Grants 71672183 and 72101241, Fundamental Research Funds for the Central University, China University of Geosciences (Wuhan), and Social Sciences and Humanities Research Council of Canada (SSHRC) Grant (Grant No. 435-2020-0761).

Appendix A Summary of SNA metrics in products recommendation network studies

No.	Attribute Dimensions	Metrics	Literature support	Research level
1	Centrality	Degree centrality (<i>DC</i>)	(Susarla, Oh, & Tan, 2012)	ego-network
		In-degree centrality (<i>InDC</i>)	(Carmi, et al., 2017; Chen, Wu, & Yoon, 2004; Goldenberg, et al., 2012; Lin, et al., 2017; Lin, 2014; Oestreicher-Singer & Sundararajan, 2012b; Pathak, et al., 2010; Stephen & Toubia, 2010; Susarla, et al., 2012) (Lin & Wang, 2018)	ego-network
		Out-degree centrality (<i>OutDC</i>)	(Goldenberg, et al., 2012; Lin, et al., 2017; Stephen & Toubia, 2010; Susarla, et al., 2012) (Lin & Wang, 2018)	dyadic-network ego-network dyadic-network
2		Closeness centrality (<i>CIC</i>)	(Goldenberg, et al., 2012)	ego-network
		In-closeness centrality (<i>InCIC</i>)	(Stephen & Toubia, 2010)	ego-network
		Out-closeness centrality (<i>OutCIC</i>)	(Stephen & Toubia, 2010)	ego-network
3		Betweenness centrality (<i>BC</i>)	(Goldenberg, et al., 2012)	ego-network
4		Eigenvector centrality (<i>EC</i>)	(Jabr & Zheng, 2014)	dyadic-network
5		In-eigenvector (authority) centrality (<i>InEC</i>)	(Stephen & Toubia, 2010)	ego-network
		Out-eigenvector (hub) centrality (<i>OutEC</i>)	(Stephen & Toubia, 2010)	ego-network
		PageRank (<i>PR</i>)	(Goldenberg, et al., 2012; Oestreicher-Singer, et al., 2013) (Oestreicher-Singer & Sundararajan, 2012a)	ego-network global-network
6	Connection	Assortative mixing (homophily) (<i>AM</i>)	(Oestreicher-Singer & Sundararajan, 2012a) (Oestreicher-Singer & Sundararajan, 2012b) (Goldenberg, et al., 2012)	global-network ego-network ego-network
7		Clustering coefficient (<i>CC</i>)	(Oestreicher-Singer & Sundararajan, 2012a) (Lin, et al., 2017) (Stephen & Toubia, 2010)	global-network ego-network ego-network
8	Scale	Incoming clustering coefficient (<i>InCC_i</i>)	(Stephen & Toubia, 2010)	ego-network
		Outgoing clustering coefficient (<i>OutCC_i</i>)	(Stephen & Toubia, 2010)	ego-network
		Size	(Oestreicher-Singer & Sundararajan, 2012a)	global-network

References

- Abbasi-Moud, Z., Hosseinabadi, S., Kelarestaghi, M., & Eshghi, F. (2022). CAFOB: Context-aware fuzzy-ontology-based tourism recommendation system. *Expert Systems with Applications*, 199, Article 116877.
- Abbasi-Moud, Z., Vahdat-Nejad, H., & Sadri, J. (2021). Tourism recommendation system based on semantic clustering and sentiment analysis. *Expert Systems with Applications*, 167, Article 114324.
- Adamopoulos, P., Ghose, A., & Tuzhilin, A. (2022). Heterogeneous demand effects of recommendation strategies in a mobile application: Evidence from econometric models and machine-learning instruments. *MIS Quarterly*, 45(1), 101–150.
- Adomavicius, G., Bockstedt, J. C., Curley, S. P., & Zhang, J. (2013). Do recommender systems manipulate consumer preferences? A study of anchoring effects. *Information Systems Research*, 24(4), 956–975.
- AnalysisConsulting. (2019). Quarterly monitoring and analysis of china's online vacation travel market in Q4, 2018. In. <https://www.analysys.cn/article/detail/20019196>.
- Anderson, C. (2006). *The long tail: Why the future of business is selling less of more*. New York: Hyperion Press.
- Baum, D., & Spann, M. (2014). The interplay between online consumer reviews and recommender systems: An experimental analysis. *International Journal of Electronic Commerce*, 19(1), 129–162.
- Bobadilla, J., Ortega, F., Hernando, A., & Gutiérrez, A. (2013). Recommender systems survey. *Knowledge-Based Systems*, 46(1), 109–132.
- Brin, S., Page, L., Motwani, R., & Winograd, T. (1999). The pagerank citation ranking: Bringing order to the web. In: Stanford InfoLab.
- Brynjolfsson, E., Hu, Y. J., & Smith, M. D. (2006). From niches to riches: Anatomy of the long tail. *MIT Sloan Management Review*, 47(4), 67–71.
- Burke, R. (2002). Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction*, 12(4), 331–370.
- Carmi, E., Oestreicher-singer, G., Stettner, U., & Sundararajan, A. (2017). Is Oprah contagious? The depth of diffusion of demand shocks in a product network. *MIS Quarterly*, 41(1), 207–222.
- Chen, L., Cao, J., Wang, Y., Liang, W., & Zhu, G. (2022). Multi-view graph attention network for travel recommendation. *Expert Systems with Applications*, 191, Article 116234.
- Chen, P., Wu, S., & Yoon, J. (2004). The impact of online recommendations and consumer feedback on sales. In *the Proceedings of International Conference on Information Systems*.
- Chen, Y. H., Lu, Y. B., Wang, B., & Pan, Z. (2019). How do product recommendations affect impulse buying? An empirical study on WeChat social commerce. *Information & Management*, 56(2), 236–248.
- Choi, I. Y., Oh, M. G., Kim, J. K., & Ryu, Y. U. (2016). Collaborative filtering with facial expressions for online video recommendation. *International Journal of Information Management*, 36(3), 397–402.
- Cosley, D., Lam, S. K., Albert, I., Konstan, J. A., & Riedl, J. (2003). Is seeing believing?: How recommender system interfaces affect users' opinions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 585–592).
- Fleder, D., & Hosanagar, K. (2009). Blockbuster culture's next rise or fall: The impact of recommender systems on sales diversity. *Management Science*, 55(5), 697–712.
- Franceschet, M. (2011). Pagerank: Standing on the shoulders of giants. *Communications of the ACM*, 54(6), 92–101.
- Ghasemaghahi, M., Hassanein, K., & Benbasat, I. (2019). Assessing the design choices for online recommendation agents for older adults: Older does not always mean simpler information technology. *MIS Quarterly*, 43(1), 329–346.
- Goldenberg, J., Oestreichersinger, G., & Reichman, S. (2012). The quest for content: How user-generated links can facilitate online exploration. *Journal of Marketing Research*, 49(4), 452–468.
- Gorgoglione, M., Panniello, U., & Tuzhilin, A. (2019). Recommendation strategies in personalization applications. *Information & Management*, 56(6), Article 103143.
- Häubl, G., & Murray, K. B. (2003). Preference construction and persistence in digital marketplaces: The role of electronic recommendation agents. *Journal of Consumer Psychology*, 13(1–2), 75–91.
- Hair, J., Anderson, R., Tatham, R., & Black, W. (2010). *Multivariate data analysis* (7th ed.). NJ: Prentice Hall.
- Han, K., Kauffman, R. J., & Nault, B. R. (2011). Returns to information technology outsourcing. *Information Systems Research*, 22(4), 824–840.
- Handarkho, Y. D. (2021). Social experience vs. social technology in enhancing the intention to use social commerce: A case study of Indonesia. *Journal of Enterprise Information Management*, 34(3), 860–883.
- Hanneman, R. A., & Riddle, M. (2011). Concepts and measures for basic network analysis. In J. Scott & P. J. Carrington (Eds.), *The SAGE Handbook of Social Network Analysis* (pp. 340–369).
- He, J., & Chu, W. W. (2010). A social network-based recommender system. *Data Mining for Social Network Data* (1st, pp. 47–74). Boston, MA: Springer.
- He, J., Liu, G., Tan, X., & Li, T. (2021). Research on collaborative recommendation of dynamic medical services based on cloud platforms in the industrial interconnection environment. *Technological Forecasting and Social Change*, 170, Article 120895.
- Hinz, J. O., & Eckert, D. K. J. (2010). The impact of search and recommendation systems on sales in electronic commerce. *Business & Information Systems Engineering*, 2(2), 67–77.
- Hosanagar, K., Krishnan, R., & Ma, L. (2008). Recommended for you: The impact of profit incentives on the relevance of online recommendations. In *Proceedings of International Conference on Information Systems (2008)*, Article 31. <https://aisel.aisnet.org/icis2008/31>.
- Huang, H. J., Yang, J., & Zheng, B. (2021). Demand effects of product similarity network in e-commerce platform. *Electronic Commerce Research*, 21(2), 297–327.
- Im, I., Jun, J., Oh, W., & Jeong, S. O. (2016). Deal-seeking versus brand-seeking: Search behaviors and purchase propensities in sponsored search platforms. *MIS Quarterly*, 40(1), 187–203.
- Jabr, W., & Zheng, E. (2014). Know yourself and know your enemy: An analysis of firm recommendations and consumer reviews in a competitive environment. *MIS Quarterly*, 38(3), 635–654.
- Jannach, D., & Jugovac, M. (2019). Measuring the business value of recommender systems. *ACM Transactions on Management Information Systems (TMIS)*, 10(4), 1–23.
- Jing, N., Jiang, T., Du, J., & Sugumaran, V. (2018). Personalized recommendation based on customer preference mining and sentiment assessment from a Chinese e-commerce website. *Electronic Commerce Research*, 18(1), 159–179.
- Khemani, R. S., & Shapiro, D. M. (1993). *Glossary of industrial organisation economics and competition law*. France Paris: Directorate for Financial, Fiscal and Enterprise Affairs, OECD.
- Lee, D., & Hosanagar, K. (2020). How do product attributes and reviews moderate the impact of recommender systems through purchase stages? *Management Science*, 67(1), 524–546.
- Leem, B., & Chun, H. (2014). An impact of online recommendation network on demand. *Expert Systems with Applications*, 41(4), 1723–1729.
- Li, L., Chen, J., & Raghunathan, S. (2018). Recommender system rethink: Implications for an electronic marketplace with competing manufacturers. *Information Systems Research*, 29(4), 1003–1023.
- Lin, Z., Goh, K. Y., & Heng, C. S. (2017). The demand effects of product recommendation networks: An empirical analysis of network diversity and stability. *MIS Quarterly*, 41(2), 397–426.
- Lin, Z., & Wang, Q. (2018). E-commerce product networks, word-of-mouth convergence, and product sales. *Journal of the Association for Information Systems*, 19(1), 23–39.
- Lin, Z. J. (2014). An Empirical Investigation of User and System Recommendations in E-commerce. *Decision Support Systems*, 68, 111–124.
- Liu, Q. H., Zhang, X. Y., Zhang, L. Y., & Zhao, Y. (2019). The interaction effects of information cascades, word of mouth and recommendation systems on online reading behavior: An empirical investigation. *Electronic Commerce Research*, 19(3), 521–547.
- Lu, J., Wu, D., Mao, M., Wang, W., & Zhang, G. (2015). Recommender system application developments: A survey. *Decision Support Systems*, 74, 12–32.
- Majd, E., & Hobson, M. (2020). An incorporated RUU model for multi-agent systems in e-commerce. *Journal of Enterprise Information Management*, 33(5), 905–921.
- Mitra, K., Reiss, M. C., & Capella, L. M. (1999). An examination of perceived risk, information search and behavioral intentions in search, experience and credence services. *Journal of Services Marketing*, 13(3), 208–228.
- Nelson, P. (1970). Information and consumer behavior. *Journal of Political Economy*, 78(2), 311–329.
- Newman, M. (2010). *Networks: An introduction*. London: Oxford University Press.
- Oestreicher-Singer, G., Libai, B., Sivan, L., Carmi, E., & Yassin, C. (2013). The network value of products. *Journal of Marketing*, 77(3), 1–14.
- Oestreicher-Singer, G., & Sundararajan, A. (2012a). Recommendation networks and the long tail of electronic commerce. *MIS Quarterly*, 36(1), 65–83.
- Oestreicher-Singer, G., & Sundararajan, A. (2012b). The visible hand? Demand effects of recommendation networks in electronic markets. *Management Science*, 58(11), 1963–1981.
- Otte, E., & Rousseau, R. (2002). Social network analysis: A powerful strategy, also for the information sciences. *Journal of Information Science*, 28(6), 441–453.
- Park, S. H., & Han, S. P. (2013). From accuracy to diversity in product recommendations: Relationship between diversity and customer retention. *International Journal of Electronic Commerce*, 18(2), 51–71.
- Pathak, B., Garfinkel, R., Gopal, R. D., Venkatesan, R., & Yin, F. (2010). Empirical analysis of the impact of recommender systems on sales. *Journal of Management Information Systems*, 27(2), 159–188.
- Pazzani, M. J., & Billsus, D. (2007). Content-based recommendation systems. In P. Brusilovsky, A. Kobsa, & W. Nejdl (Eds.), *The Adaptive Web: Methods and Strategies of Web Personalization* (pp. 325–341). Berlin, Heidelberg: Springer.
- Ren, F., & Dewan, S. (2015). Industry-level analysis of information technology return and risk: What explains the variation? *Journal of Management Information Systems*, 32(2), 71–103.
- Schafer, J. B., Dan, F., Herlocker, J., & Sen, S. (2007). *Collaborative filtering recommender systems*. Berlin Heidelberg: Springer.
- Shao, B., Li, X., & Bian, G. (2021). A survey of research hotspots and frontier trends of recommendation systems from the perspective of knowledge graph. *Expert Systems with Applications*, 165, Article 113764.
- Son, J., & Kim, S. B. (2017). Content-based filtering for recommendation systems using multiattribute networks. *Expert Systems with Applications*, 89, 404–412.
- Stephen, A. T., & Toubia, O. (2010). Deriving value from social commerce networks. *Journal of Marketing Research*, 47(2), 215–228.
- Susarla, A., Oh, J. H., & Tan, Y. (2012). Social networks and the diffusion of user-generated content: Evidence from youtube. *Information Systems Research*, 23(1), 23–41.
- Wang, Q., Yu, J. J., & Deng, W. W. (2019). An adjustable re-ranking approach for improving the individual and aggregate diversities of product recommendations. *Electronic Commerce Research*, 19(1), 59–79.
- Wang, W., Xu, J., & Wang, M. (2018). Effects of recommendation neutrality and sponsorship disclosure on trust vs. Distrust in online recommendation agents: Moderating role of explanations for organic recommendations. *Management Science*, 64(11), 5198–5219.
- Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data (2nd edition)*. Cambridge, MA: MIT Press.

- Xiao, J. H., Wu, Y., Xie, K., & Hu, Q. (2019). Managing the e-commerce disruption with IT-based innovations: Insights from strategic renewal perspectives. *Information & Management*, 56(1), 122–139.
- Xu, J. J., Benbasat, I., & Cenfetelli, R. T. (2014). The nature and consequences of trade-off transparency in the context of recommendation agents. *MIS Quarterly*, 38(2), 379–406.
- Xu, Y., Ma, J., Sun, Y. H., Hao, J., Sun, Y., & Zhao, Y. (2009). Using social network analysis as a strategy for e-commerce recommendation. *Proceeding of Pacific Asia Conference on Information Systems*.
- Yang, D., & Gao, X. (2017). Online retailer recommender systems: A competitive analysis. *International Journal of Production Research*, 55(14), 4089–4109.
- Yang, N., Jo, J., Jeon, M., Kim, W., & Kang, J. (2022). Semantic and explainable research-related recommendation system based on semi-supervised methodology using BERT and IDA models. *Expert Systems with Applications*, 190, Article 116209.
- Zhang, H., Zhao, L., & Gupta, S. (2018). The role of online product recommendations on customer decision making and loyalty in social shopping communities. *International Journal of Information Management*, 38(1), 150–166.
- Zhang, M. Y., & Bockstedt, J. (2020). Complements and substitutes in online product recommendations: The differential effects on consumers' willingness to pay. *Information & Management*, 57(6), Article 103341.
- Zhang, Z., Lin, H., Liu, K., Wu, D., Zhang, G., & Lu, J. (2013). A hybrid fuzzy-based personalized recommender system for telecom products/services. *Information Sciences*, 235(6), 117–129.
- Zhu, Z., Lin, S., Jiang, Y., & Liu, Q. (2022). The coordination effect of B2B digital process capabilities on competitive performance: balancing or complementing. *Journal of Enterprise Information Management*, 35(3), 918–946.
- Zhu, Z., Zhao, J., & Bush, A. A. (2020). The effects of e-business processes in supply chain operations: Process component and value creation mechanisms. *International Journal of Information Management*, 50, 273–285.

Further reading

- Lin, Z., Goh, K. Y., & Heng, C. S. (2017). The demand effects of product recommendation networks: An empirical analysis of network diversity and stability. *MIS Quarterly*, 41(2), 397–426.
- Oestreicher-Singer, G., & Sundararajan, A. (2012). Recommendation networks and the long tail of electronic commerce. *MIS Quarterly*, 36(1), 65–83.